

Self-Correcting Multimodal AI Agent for Automated Expense Report Generation from Receipt Images

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I. Problem Summary

Extracting structured information from receipt images is a complex challenge in computer vision and artificial intelligence. Real-world receipts - particularly in diverse environments like India - present high variability in layouts, fonts, and languages. Furthermore, receipt images suffer from physical degradation, skewed angles, low lighting, and noise, which severely degrade the performance of conventional Optical Character Recognition (OCR) systems.

Traditional single-pass OCR and Vision-Language Models (VLMs) frequently misinterpret densely packed text or hallucinate values, leading to critical errors in financial reporting. A single misclassification of a tax value or decimal placement can render an entire expense extraction invalid. Currently, most automated expense systems rely on static, rule-based extraction pipelines that lack the cognitive reasoning required to detect their own errors.

The core AI challenge is not just perception (reading the text), but reasoning (understanding the relationships between extracted entities). Information such as line items, subtotals, and taxes inherently follow mathematical constraints (e.g., $\text{Subtotal} + \text{Tax} = \text{Total}$). A sophisticated AI system must leverage these constraints to perform agentic self-verification. By modeling the extraction process as an iterative, self-correcting agent rather than a linear feed-forward pipeline, the system can detect contradictions, re-examine uncertain image regions, and dynamically correct its own inferences before finalizing structured output.

This self-correcting capability fundamentally shifts the paradigm from passive data extraction to active problem-solving. When the verification engine detects that the extracted numbers do not arithmetically align, a conventional system would either crash, require manual human intervention, or worse, silently output erroneous data. In contrast, the proposed self-correcting agent treats this mathematical failure as an active trigger for a directed feedback loop. It formulates an internal hypothesis regarding the source of the error - such as identifying a specific spatial region where a character might have been misread due to a crease in the paper or recognizing that a secondary local tax line was completely missed during the initial pass.

Instead of restarting the entire extraction from scratch, the agent autonomously re-prompts the vision-language module, directing its "attention" specifically to these high-uncertainty regions. It essentially asks targeted questions of the image (e.g., *"The subtotal and tax sum to 45000, but the extracted total is 40000. Re-evaluate the tax field for missing decimals."*). The system then compares the newly extracted values against the rigid mathematical constraints. This continuous cycle of error detection, hypothesis generation, targeted re-examination, and mathematical validation mimics human cognitive behavior when double-checking a complex document. Ultimately, this ensures that the AI only commits data to the final expense report when it is not just perceived but mathematically proven to be consistent.

II. Discussion on related works :

S. No.	Paper/Patent (Year, author)	Summary	Methodology	Observations	Gaps / Limitations
1.	ICDAR 2019 SROIE (2019, Huang et al.)	Introduced a benchmark dataset for receipt extraction.	Evaluated OCR and sequence labeling on scanned receipts.	High baseline accuracy for clean receipts.	Did not account for heavily wrinkled or real-world noisy images.
2.	CORD (2019, Park et al.)	Presented a consolidated receipt dataset for spatial extraction.	Annotated complex Indonesian receipts for line-item parsing.	Emphasized that spatial layout is critical for parsing.	Lacked reasoning mechanisms for self-correction.
3.	LayoutLM (2020, Xu et al.)	Pre-training method integrating text and layout.	Jointly modeled text and 2D bounding box coordinates using Transformers.	Achieved state-of-the-art on form understanding.	Relied heavily on external OCR engines.
4.	LayoutLMv2 (2021, Xu et al.)	Multimodal pre-training for documents.	Integrated image embeddings with text and layout in a single architecture.	Outperformed text-only models in visually rich documents.	Single-pass inference; no error recovery.
5.	StrucText (2021, Li et al.)	Structured text understanding via segment-level representation.	Used self-supervised pre-training tasks for segment grouping.	Excellent entity linking performance.	Did not apply arithmetic verification on extracted entities.

6.	Donut (2022, Kim et al.)	OCR-free document understanding model.	End-to-end Transformer mapping images directly to JSON.	Reduced compounding errors from poor OCR.	Prone to hallucination on long line items without constraints.
7.	ReAct (2023, Yao et al.)	Framework combining reasoning and acting in LLMs.	Prompted LLMs to generate reasoning traces before acting.	Significantly improved task completion and reduced hallucination.	Not specifically tailored for dense document verification.
8.	Reflexion (2023, Shinn et al.)	Verbal reinforcement learning for agents.	Agents reflect on past failures to iteratively improve future outputs.	Improved performance across reasoning benchmarks.	Evaluated mostly on text, lacking multimodal visual integration.
9.	Self-Refine (2023, Madaan et al.)	Iterative refinement of LLM outputs via self-feedback.	LLM evaluates its own output and generates a refined version.	Demonstrated consistent improvements without fine-tuning.	Lacks mechanisms to re-evaluate the source image upon detecting errors.
10.	DocLLM (2024, Wang et al.)	Spatial layout modeling natively within LLMs.	Disentangled spatial coordinates from text during pre-training.	Better handled irregular layouts and forms.	Remained a single-pass generative model without closed-loop verification.

III. Outcome on the review:

1. Gap / Issue identified and our proposal

Gap: Current multimodal document models extract data in a single forward pass and cannot detect or recover from logical or arithmetic errors (e.g., extracting a tax amount that contradicts the total).

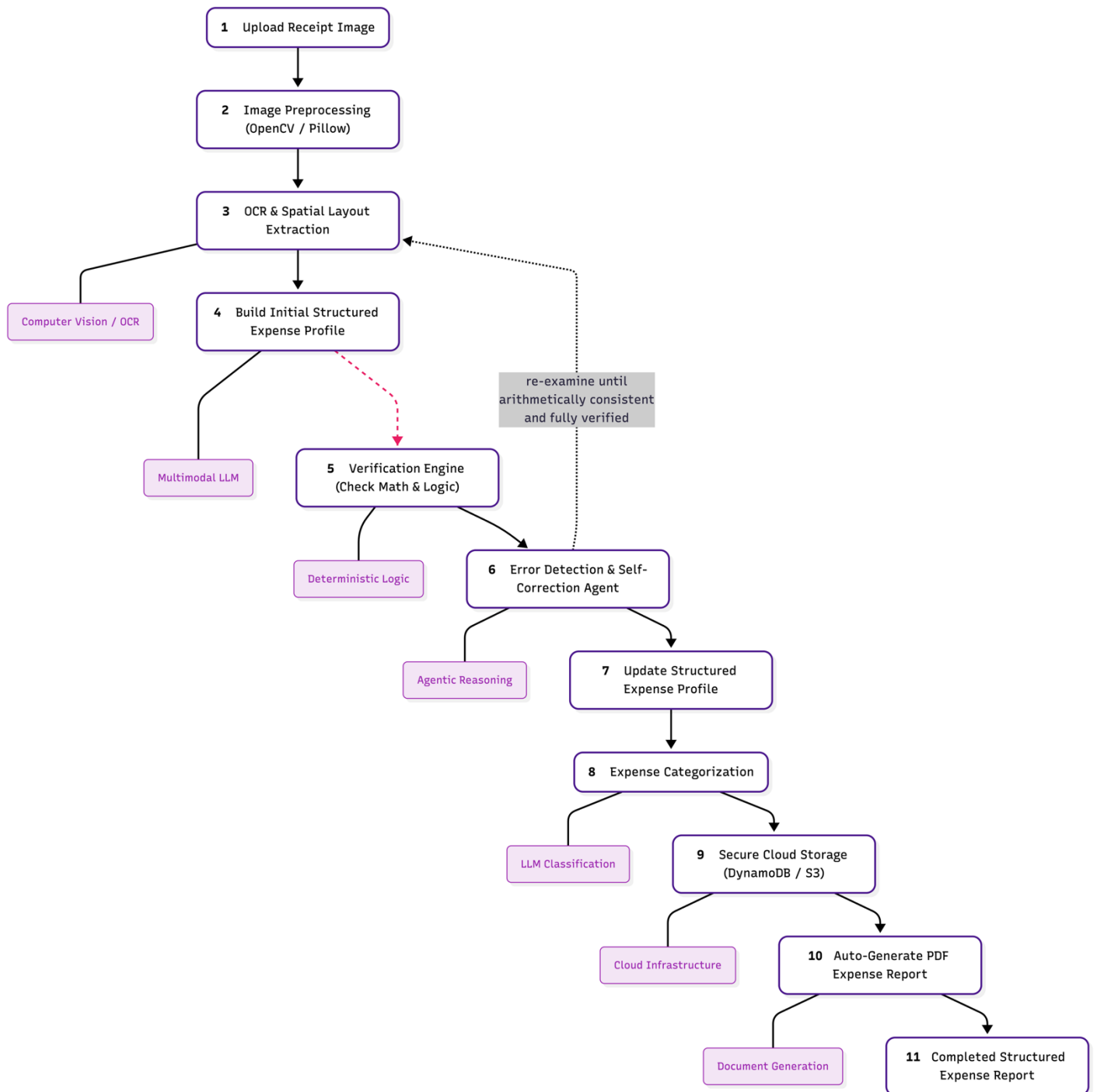
Proposal: We propose an agentic, multimodal AI framework that combines image processing, text extraction, and a dedicated self-correction feedback loop. The agent actively verifies arithmetic consistency and semantic logic, iteratively re-prompting the vision-language module to re-examine uncertain regions until a highly confident, verified extraction is achieved.

2. Clear statement of WHAT'S NEW in this proposal

The novelty lies in applying agentic decision-making and self-correction directly to structured multimodal document extraction. While iterative reasoning exists in pure text LLMs, this project introduces a self-correcting multimodal workflow that treats arithmetic consistency (e.g., verifying that line items sum to the total) as an intrinsic verification signal, allowing the AI to autonomously decide when to re-process an image before finalizing the output.

IV. Framework

1. Architecture diagram



2. Details (Explain the modules/blocks one by one)

1. Upload Receipt Image:

- **Purpose:** Ingests the raw, unstructured receipt image into the system. This serves as the initial input for the AI pipeline, capturing documents that may have varying layouts, lighting conditions, and orientations.

2. Image Preprocessing (OpenCV / Pillow):

- **Purpose:** Enhances and normalizes the uploaded image to optimize it for machine perception.
- **Processing:** Applies computer vision techniques such as adaptive thresholding, noise reduction, and deskewing to ensure the text is as clear as possible before extraction.

3. OCR & Spatial Layout Extraction (Computer Vision / OCR):

- **Purpose:** Forms the foundational perception layer of the system.
- **Processing:** Extracts raw text and maps the 2D spatial bounding-box coordinates of every detected word. This spatial awareness is critical for understanding the hierarchical layout of complex receipts.

4. Build Initial Structured Expense Profile (Multimodal LLM):

- **Purpose:** Translates unstructured visual and textual data into a preliminary structured format (e.g., JSON).
- **Processing:** The LLM interprets the spatial relationships between text blocks to identify key entities such as the vendor name, purchase date, individual line items, tax amounts, and the grand total.

5. Verification Engine (Check Math & Logic) (Deterministic Logic):

- **Purpose:** Acts as the mathematical reasoning core of the system.
- **Processing:** Automatically applies deterministic constraints to the extracted profile. It calculates whether the sum of the line items plus the extracted tax matches the extracted total amount, and flags any missing mandatory fields.

6. Error Detection & Self-Correction Agent (Agentic Reasoning):

- **Purpose:** Drives the autonomous self-correction loop, distinguishing this system from traditional single-pass OCR pipelines.
- **Processing:** If the Verification Engine flags an arithmetic mismatch or missing data, this agent formulates a targeted re-evaluation prompt (e.g., "The extracted tax of ₹500.00 and subtotal of ₹4000.00 do not equal the total of

₹5000.00. Re-examine the tax field."). It then loops back to step 3/4 to re-process the uncertain regions of the image until mathematical consistency is achieved.

7. Update Structured Expense Profile:

- **Purpose:** Finalizes the data structure.
- **Processing:** Once the self-correction loop successfully resolves all inconsistencies, this module locks in the verified data, ensuring that only highly accurate information moves forward in the pipeline.

8. Expense Categorization (LLM Classification):

- **Purpose:** Assigns semantic meaning to the verified transaction.
- **Processing:** Uses an LLM to analyze the vendor name and line items to autonomously classify the overall expense into standard accounting categories (e.g., Travel, Meals, Office Supplies).

9. Secure Cloud Storage (DynamoDB / S3) (Cloud Infrastructure):

- **Purpose:** Provides scalable and secure data persistence.
- **Processing:** Stores the fully validated, structured JSON expense data in a NoSQL database (DynamoDB) for fast querying, while archiving the original receipt image in object storage (S3).

10. Auto-Generate PDF Expense Report (Document Generation):

- **Purpose:** Converts the digital data back into a human-readable format.
- **Processing:** Programmatically maps the verified, structured data into a standardized, professional PDF layout for accounting and auditing purposes.

11. Completed Structured Expense Report:

- **Purpose:** The final, verified output of the end-to-end AI agent workflow, ready for enterprise use or direct integration into broader financial software systems.

V. Implementation details / Experimental set-up

1. Knowledge representation

A structured expense profile (vendor, date, line items, tax, total, and known/unknown fields), a verification-state record (arithmetic consistency status, discrepancy flags), and an agentic decision-trace entry per self-correction loop - the same three structures track the receipt through the entire pipeline, whether processed locally or via AWS Lambda.

2. Extraction layer

Profile fields are pulled from preprocessed receipt images using Tesseract OCR and Multimodal LLMs. The models use spatial layout extraction to parse vendor details, individual line items, and totals, feeding the exact structured profile that the reasoning and verification stages consume.

3. The self-correction decision function

Re-evaluation is triggered based on mathematical consistency and extraction completeness. If an arithmetic mismatch occurs (e.g., $\text{Subtotal} + \text{Tax} \neq \text{Total}$), the nature of the discrepancy dictates the formulation of a targeted prompt (e.g., "Re-check the tax field"). The most severe logical contradiction is queried first; the loop stops once arithmetic logic is fully satisfied or a maximum iteration limit is reached.

4. Verification-gating state machine

Each extracted receipt moves through up to four distinct states - initial extraction, awaiting mathematical verification, flagged for iterative re-examination, and fully resolved/verified. Persistent arithmetic mismatches that fail self-correction are logged as flagged for review, never silently accepted or incorrectly dropped into the final ledger.

5. Deterministic verification

Arithmetic constraints (e.g., verifying that line items sum exactly to the final amount) run as a pure deterministic logic function on the resolved JSON profile, with no dependency on the LLM's generative probabilities. This maintains a clean separation between the probabilistic text extraction and the rigid mathematical rules required for accounting.

6. Baseline comparison and validation

The proposed experimental setup will evaluate the system using the SROIE and CORD datasets. The self-correcting agent will be explicitly tested against a baseline single-pass extraction model. On sample complex receipts, the experiment will track whether the iterative reasoning loop correctly identifies its own initial errors (such as hallucinated tax values) and successfully amends them to produce higher field-level extraction accuracy.

7. Behavioural testing against live input

Run end-to-end against real-world, noisy Indian receipt images, the engine is designed to automatically extract stated text, identify arithmetic inconsistencies caused by low image quality, iteratively prompt the vision encoder to fix the errors, securely store the resolved data in DynamoDB, and generate the final PDF expense report.